

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 14-A/B

Numerical methods for solving initial values problems

Recap We've covered numerical methods for solving *algebraic* equations.

- Single non-linear equations such as $x^4 - 3x^3 + x^2 + 1 = 0$

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- Systems of non-linear equations such as $\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$

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- Systems of linear equations such as the "forces in a crane" problem.

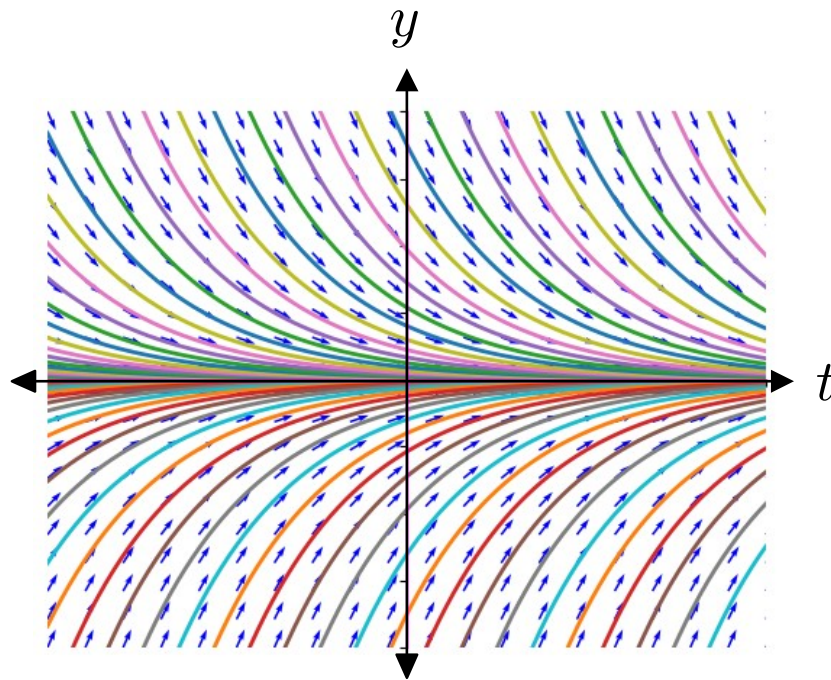
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Ordinary differential equation (ODE)

An ODE is an equation whose unknown is a ***function*** $y : \mathbb{R} \rightarrow \mathbb{R}$

Example Find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that $y' = -y$

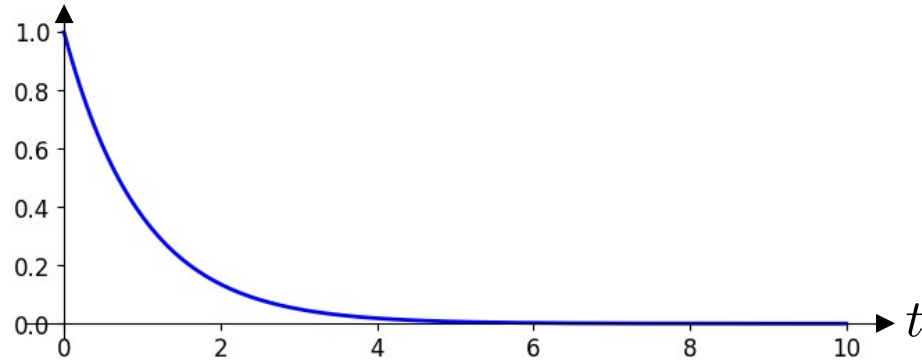
- Solution: $y(t) = a e^{-t}$ $a \in \mathbb{R}$



Initial value problem (IVP)

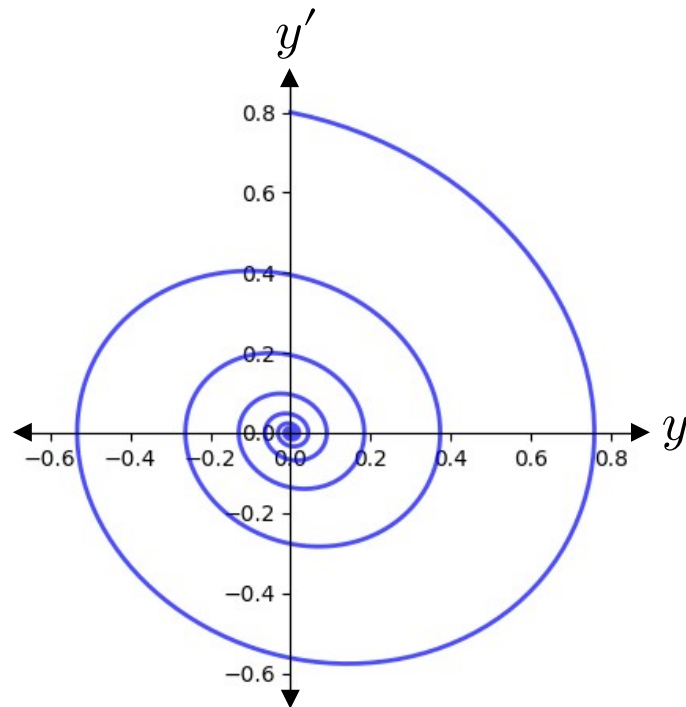
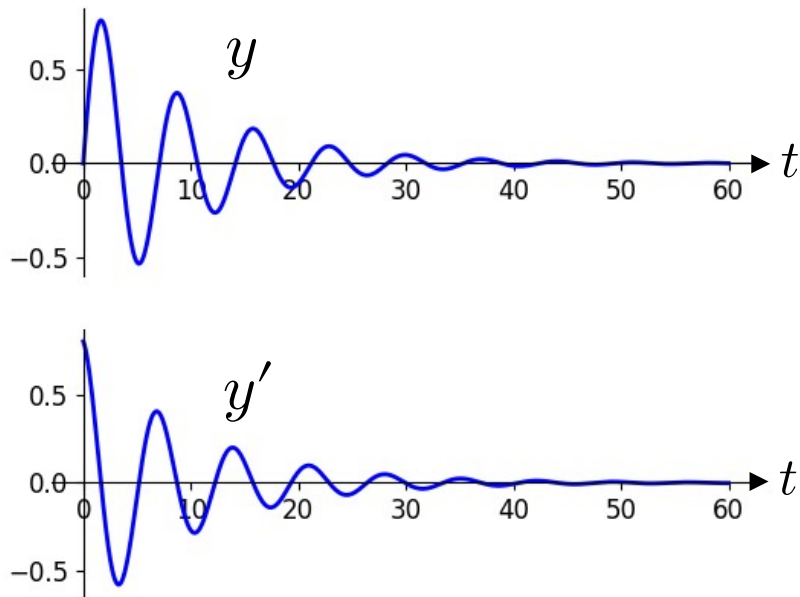
An IVP is an ODE with the value of y at time zero given.

Example $y' = -y$ with $y(0) = 1$



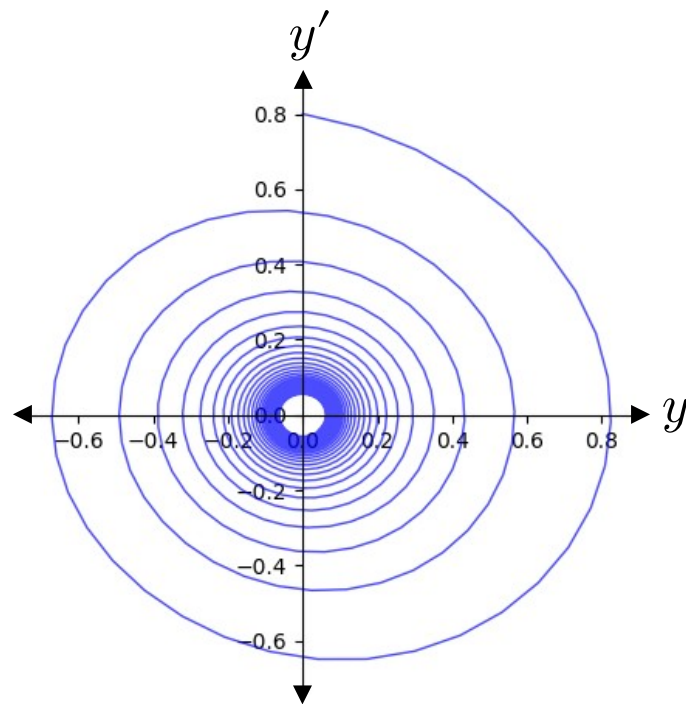
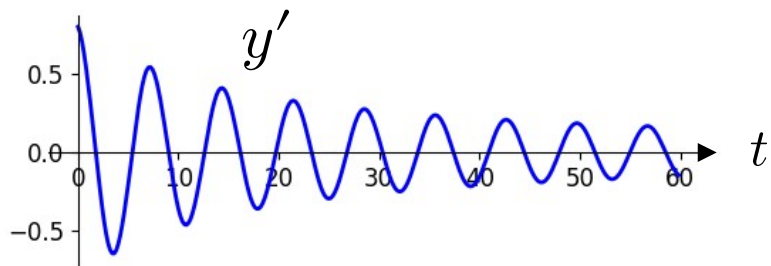
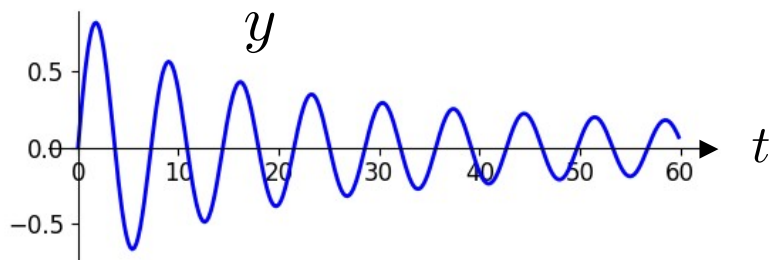
Example Given $a > 0$, $b > 0$, find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that,

$$y'' + a y' + b y = 0$$



Example Given $a > 0$, $b > 0$, find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that,

$$y'' + a y' |y'| + b \sin(y) = 0$$

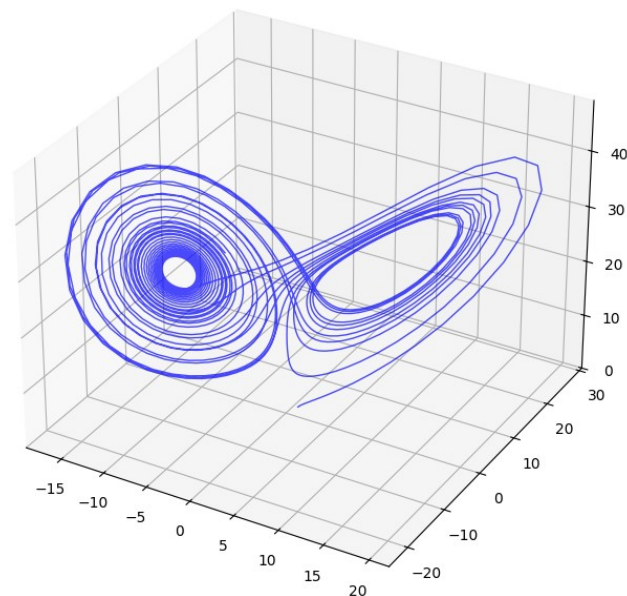
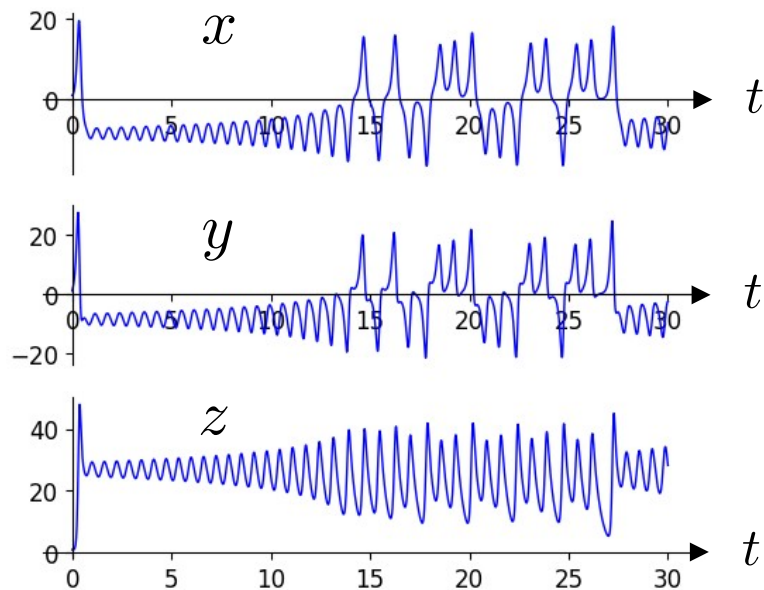


Example Given $\sigma > 0$, $\rho > 0$, $\beta > 0$, find scalar functions $x(t)$, $y(t)$, $z(t)$ such that

$$x' = \sigma(y - x)$$

$$y' = x(\rho - z) - y$$

$$z' = xy - \beta z$$



Solving an IVP numerically

1) If the ODE is of order >1 , put it into **vector form**: $\mathbf{y}' = f(\mathbf{y}, t)$

2) Apply a **numerical method** to the vector form.

- Euler's method
- Midpoint method
- Heun's method
- etc.

Putting an ODE into vector form $\mathbf{y}' = f(\mathbf{y}, t)$

Example: $y'' + a y' + b y = 0$

1) Define n new variables representing y and its first $n - 1$ derivatives.

2) $y'_i = y_{i+1}$ for each i from 1 to $n-1$.

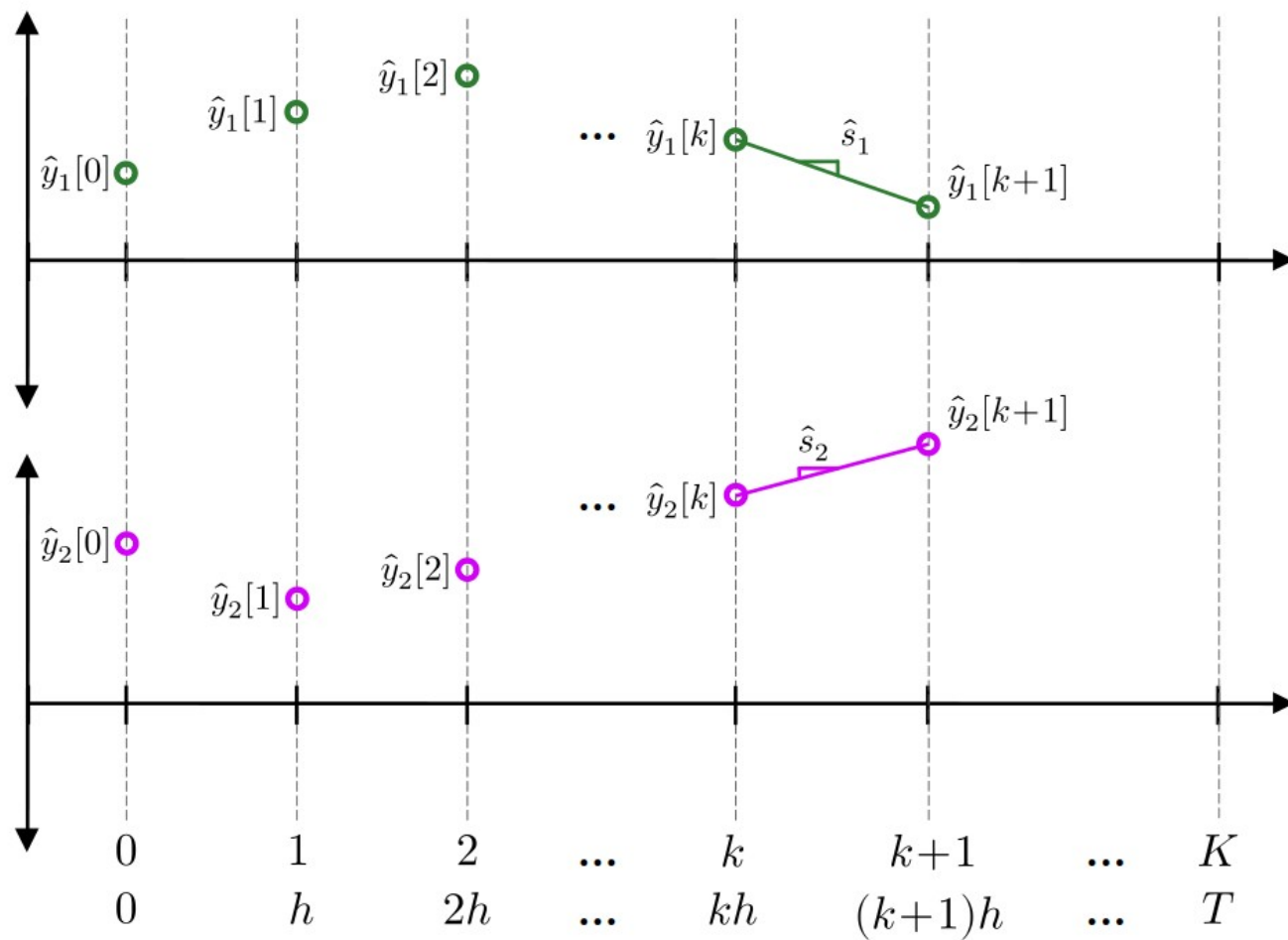
3) Use the ODE to express y'_n in terms of y_1 through y_n .

4) Write the vector form $\mathbf{y}' = f(\mathbf{y}, t)$ with $\mathbf{y} = (y_1, \dots, y_n)$.

Numerical solution of $\mathbf{y}' = f(\mathbf{y}, t)$

- Generic iteration:
$$\hat{\mathbf{y}}[0] = \mathbf{y}(0)$$
$$\hat{\mathbf{y}}[k + 1] = \hat{\mathbf{y}}[k] + \hat{\mathbf{s}} h$$
- $\hat{\mathbf{y}}[k] = (\hat{y}_1[k], \dots, \hat{y}_n[k])$... numerical solution at time step k .
- $\hat{\mathbf{s}} = (\hat{s}_1, \dots, \hat{s}_n)$... vector of slopes.
- Each numerical method computes $\hat{\mathbf{s}}$ differently, based on:
 - f ... the vector state equation
 - h ... the time step
 - $\hat{\mathbf{y}}[k]$... the current estimate
 - t ... the current time

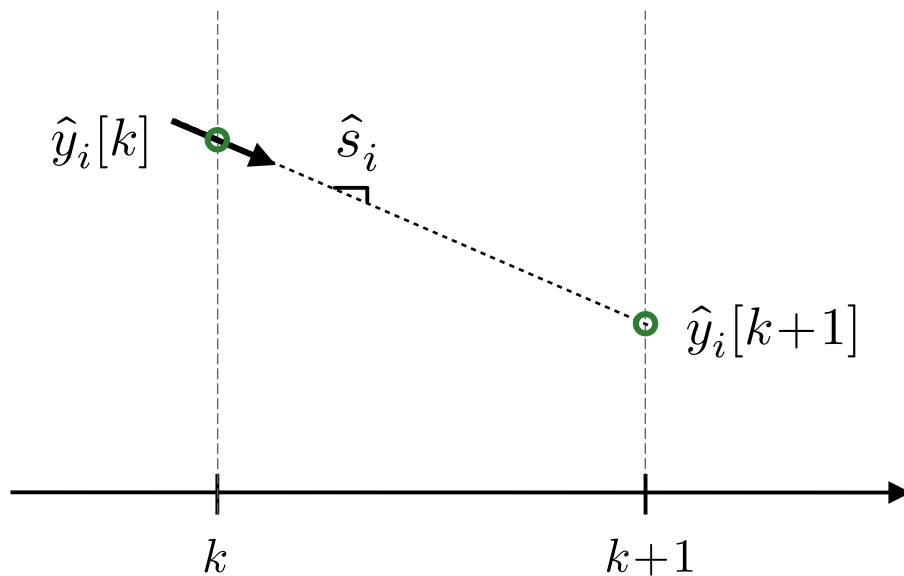
Generic iteration



Euler's method

$$\hat{\mathbf{s}} = f(\hat{\mathbf{y}}[k], t)$$

Plot of i 'th component:



Midpoint method

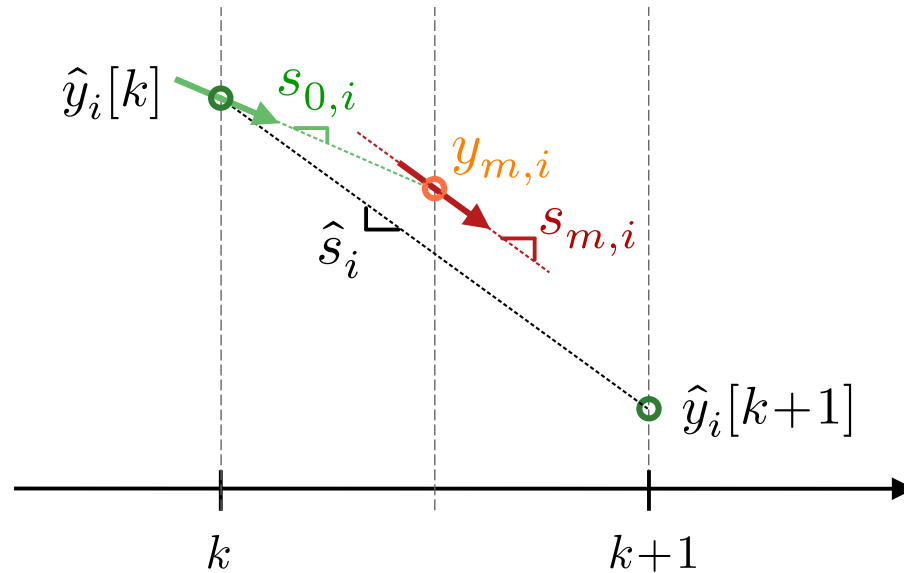
$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_m = \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0$$

$$\mathbf{s}_m = f(\mathbf{y}_m, t + h/2)$$

$$\hat{\mathbf{s}} = \mathbf{s}_m$$

Plot of i 'th component:



Heun's method

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + h \mathbf{s}_0$$

$$\mathbf{s}_1 = f(\mathbf{y}_1, t+h)$$

$$\hat{\mathbf{s}} = 0.5(\mathbf{s}_0 + \mathbf{s}_1)$$

Plot of i 'th component:

